

SECTION 8

HANDLING, SERVICING, AND MAINTENANCE

TABLE OF CONTENTS

	Page
8.1 GENERAL	8-1
8.2 GROUND HANDLING	8-1
8.2.1 Towing	8-1
8.2.2 Parking	8-2
8.2.3 Mooring	8-4
8.3 SERVICING	8-4
8.3.1 Refueling	8-4
8.3.2 Defueling	8-5
8.3.3 Engine Oil System	8-6
8.3.4 Main Transmission	8-6
8.3.5 Rotor Hub Reservoir	8-6
8.3.6 Intermediate Gearbox	8-6
8.3.7 Tail Rotor Gearbox	8-6
8.3.8 Hydraulic Systems	8-6
8.3.9 Engine Compressor Rinsing	8-7
8.4 COLD WEATHER OPERATION - GENERAL PRACTICES	8-7
8.4.1 Fuels	8-7
8.4.2 Fuel/Additive Blending	8-7
8.4.3 Mixing Alternate Fuel	8-8
8.4.4 Oils	8-9
8.4.5 Preheating	8-9
8.5 MAINTENANCE	8-9

LIST OF FIGURES

Fig. 8-1 Ground Handling	8-3
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SECTION 8

HANDLING, SERVICING, AND MAINTENANCE

8.1 GENERAL

This section describes ways an operator can ensure that the necessary handling, servicing and maintenance of the helicopter are accomplished.

NOTE It is the operator's responsibility to ensure that all airworthiness directives are complied with and that the handling, servicing and maintenance of the helicopter are accomplished when required and in accordance with the applicable Federal Aviation Regulations.

In order to meet the above requirements, the helicopter operator should establish contact with the helicopter manufacturer or certified service station for service and information; and that all correspondence regarding the helicopter include the helicopter serial number found on the data plate secured to the RH fuselage structure. Helicopter and component maintenance manuals, and parts catalogs are available from the helicopter manufacturer.

For definition of terms, abbreviations and symbols used in this section, refer to Section 1.

8.2 GROUND HANDLING

8.2.1 Towing (see Fig. 8-1)

CAUTION TO AVOID DAMAGING GYRO INSTRUMENTS, DO NOT MOVE HELICOPTER DURING THE TIME SPAN OF 10 TO 30 MINUTES AFTER SWITCHING OFF INSTRUMENT ELECTRICAL POWER.

1. Position ground handling wheels onto skid tubes making certain that lugs on skid tubes engage with flanges on wheel units.
2. Push downwards on latch lever and check that latches are locked over lugs.
3. Stabilize helicopter during raising/lowering and towing operations by grasping tail skid.
4. Close hydraulic jack valve and, using pump handle, raise helicopter.

NOTE On jacks fitted with a safety stop, lower safety stop against jack housing, then slowly open jack valve.

CAUTION HANDLE HELICOPTER USING ONLY THE FOLLOWING AUTHORIZED HARD POINTS:

- TAIL SKID
- REAR EDGE OF CABIN CEILING (BELOW TAIL BOOM)
- WINDSHIELD FRAME (ABOVE BATTERY ACCESS DOOR)

5. Maneuver helicopter into desired position.

NOTE To ease handling, maneuver helicopter in a rearward direction.

CAUTION TOW HELICOPTER AT SPEEDS LESS THAN 10 KM/HOUR. MINIMUM TURNING RADIUS WHEN TOWING IS 10 METERS.

6. Secure towbar to forward end of each skid tube and using guide, tow helicopter as required.

7. Upon reaching desired area, remove towbar.

8. Slowly open jack valve and lower helicopter to ground.

NOTE On jacks fitted with a safety stop, close jack valve and, using pump handle, release safety stop. Raise safety stop then slowly open jack valve to lower helicopter to ground.

9. Pull latch lever and remove ground handling wheels.

8.2.2 Parking

NOTE For parking periods over 3 hours or when winds are above 15 kt, main and tail rotors should be secured as a minimum.

1. Connect electrical grounding cable.

2. Make certain that all electrical power is switched off.

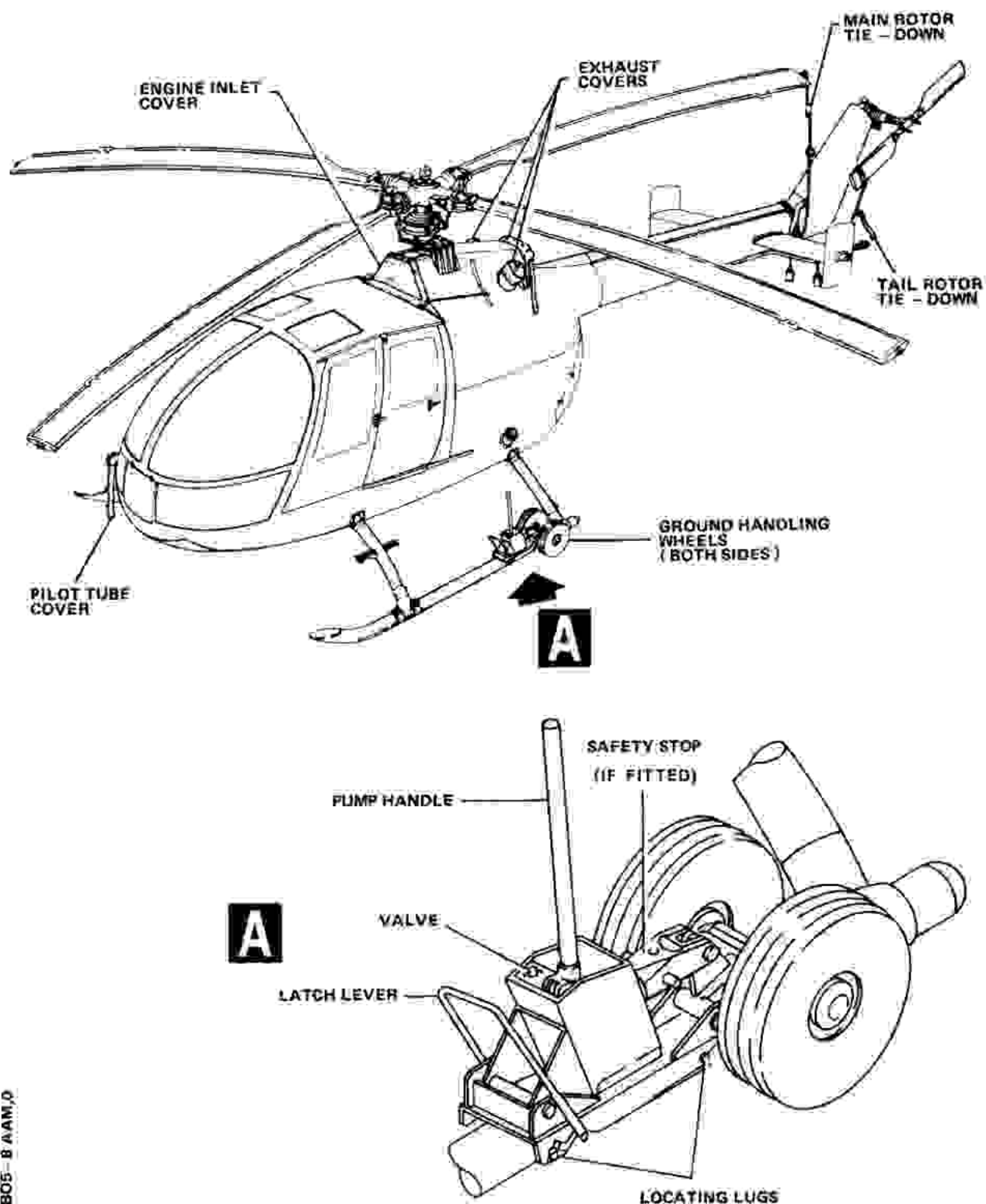
NOTE Disconnect and, if necessary, remove battery when helicopter is to be parked for long periods.

3. Secure main and tail rotors using tie-downs provided.

4. Install engine inlet, exhaust and pitot tube protective covers.

NOTE Tie-downs and protective covers are part of the helicopter flyaway kit.

5. Make certain that all doors and windows are closed and that all access panels and cowlings are secured.



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Fig. 8-1 Ground Handling

8.2.3 Mooring

When parking the helicopter in the open during periods of forecast high winds of 50 kt or higher, the helicopter should be headed in the direction from which the forecast winds are expected and then moored (anchored) as follows:

- Secure helicopter to ramp tie-downs or mooring anchors using ropes or cables wrapped around the landing gear cross tubes at both the fuselage and skid tube junctures.
- Secure rotor blades and install protective covers.
- Fill fuel tanks to max capacity and place sand bags on the skid tubes to add weight to the helicopter.

8.3 SERVICING

8.3.1 Refueling

Park helicopter on level, even surface and perform engine shutdown. Keep fire extinguisher ready at hand.

WARNING • ALWAYS ESTABLISH ELECTRICAL GROUND CONNECTION BETWEEN HELICOPTER, FUELING VEHICLE AND EXTERNAL POWER UNIT (AS REQUIRED) BEFORE STARTING REFUELING OPERATION AND DISCONNECT ONLY WHEN OPERATION IS COMPLETED.

- DO NOT CONNECT, DISCONNECT OR REMOVE BATTERY DURING OR IMMEDIATELY FOLLOWING REFUELING.

1. Establish electrical ground connection.
2. Apply power (battery or EPU) to the helicopter electrical system and check supply tank (S) fuel quantity.
3. Unlock and remove fuel filler cap.
4. Electrically ground fuel nozzle to helicopter and then insert nozzle in filler neck.
5. Service fuel tanks to required level.

NOTE If supply tank is not full (less than 75 kg), switch on the main tank fuel pumps during refueling until supply tank is full.

6. Switch off electrical power.
7. Remove fuel nozzle and disconnect its ground cable.
8. Install fuel filler cap and check locked.
9. Remove electrical ground connection.

8.3.2 Defueling

For defueling purposes, the helicopter fuel pumps are used. An external power unit should be used whenever possible to conserve battery power. Position external power unit at a safe distance (approx 10m) and connect to helicopter. If a container is used to collect the fuel during the defueling operation, that container must be electrically grounded. Keep a fire extinguisher ready at hand.

WARNING ■ ALWAYS ESTABLISH ELECTRICAL GROUND CONNECTION BETWEEN HELICOPTER, DEFUELING VEHICLE, CONTAINERS AND EXTERNAL POWER UNIT (AS REQUIRED) BEFORE STARTING DEFUELING OPERATION AND DISCONNECT ONLY WHEN OPERATION IS COMPLETED.

- DO NOT CONNECT, DISCONNECT OR REMOVE BATTERY DURING OR IMMEDIATELY FOLLOWING DEFUELING:

1. Establish electrical ground connection.
 2. Release latches and open LH/RH engine compartment access panels.
 3. Disconnect fuel supply line at the engine deck coupling of one or both engines.
 4. Connect defueling line(s) of suitable length to the coupling(s) and route free-end of line to defueling vehicle or container.
 5. Power helicopter electrical system (battery or EPU) and switch on respective fuel pumps.
- NOTE** If EPU is not available, operate only one fuel pump at a time alternating between supply and main tank pumps.
6. Monitor fuel quantity indicator. When both tanks are emptied switch off fuel pumps and de-energize helicopter electrical system.
 7. Disconnect and remove defueling line(s) and connect fuel supply line(s).
 8. Close LH/RH engine compartment access panels.
 9. Position containers and open fuel tank drain valves.
 10. Disconnect electrical ground connection.
 11. Remove containers and dispose of fuel accordingly.

8.3.3 Engine Oil System

Both engine oil tanks are located in the RH transmission compartment beneath the oil cooler assembly. The left engine oil tank is located forward of the right engine oil tank. Each tank is serviced through a gravity filler neck having a screen-type filter. An oil level sight gage is located on the outer face of each tank. When the oil level falls to the ADD OIL mark, replenish with correct oil (see Section 2) until the level reaches the MAX mark on the sight gage or until an internal float blocks off the filler neck opening.

8.3.4 Main Transmission

Access to the main transmission oil filler neck is best achieved from the engine air inlet at the cabin top. The oil level sight gage is visible from the RH transmission compartment. When the oil level falls to the MIN mark, replenish with correct oil (see Section 2) until the level reaches the MAX mark on the sight gage.

8.3.5 Rotor Hub Reservoir

The oil level of the transparent rotor hub reservoir must be between the MIN and MAX markings when viewed from the nose of the helicopter. Replenishing instructions are in the BO 105 Maintenance Manual.

8.3.6 Intermediate Gearbox

The oil level sight gage for the intermediate gearbox is viewed on the left side at the base of the vertical fin fairing. The oil level is full when the bottom half of the sight gage is filled. The oil filler cap is accessible after opening a flap above and to the right of the sight gage. Replenish with correct oil (see Section 2).

NOTE When the sight gage glass is stained, the oil level may be best verified by grasping the tail skid and shortly shaking the tail boom while carefully observing the sight gage.

8.3.7 Tail Rotor Gearbox

The oil level sight gage for the tail rotor gearbox is located on the aft end of the gearbox housing. The oil level is full when the bottom half of the sight gage is filled. The oil filler cap is located above the sight gage. Replenish with correct oil (see Section 2).

NOTE When the sight gage glass is stained, the oil level may be best verified by grasping the tail skid and shortly shaking the tail boom while carefully observing the sight gage.

8.3.8 Hydraulic Systems

The hydraulic systems should only be serviced with approved fluids (see Section 2). Since the systems are of the closed-circuit type, only trained personnel are authorized to replenish the fluids following servicing instructions in the BO 105 Maintenance Manual.

8.3.9 Engine Compressor Rinsing

- CAUTION**
- SALT-LADEN HUMIDITY AND CHEMICALS WILL CORRODE COMPRESSOR BLADES AND VANES.
 - SALT-LADEN AIR MAY BE ENCOUNTERED UP TO 280 KM INLAND UNDER CERTAIN WEATHER CONDITIONS. IF ANY DOUBT EXISTS ABOUT THE CONDITIONS IN WHICH THE ENGINES WERE OPERATED, PERFORM A DAILY COMPRESSOR WATER RINSE.

Engines subjected to salt water contamination or other chemically laden atmosphere (including pesticides and herbicides) shall undergo water rinsing after shutdown following the last flight of the day.

The rinse operation (see BO 105 Maintenance Manual) should be performed as soon as practical after flight, but not before the engines have cooled to near ambient temperature.

8.4 COLD WEATHER OPERATION - GENERAL PRACTICES

8.4.1 Fuels

When operating in ambient temperatures below +4 °C (+40 °F) the fuel must contain an anti-icing additive meeting the requirements of MIL-I-27686.

If available use JP-4 or commercial Jet B fuel. As an alternate use an AVGAS/Jet fuel mixture following limitations described in Section 2, Fuel limitations.

8.4.2 Fuel/Additive Blending

- CAUTION** IF IT CANNOT BE DETERMINED THAT THE FUEL IN THE HELICOPTER CONTAINS AN ANTI-ICING ADDITIVE OR IF THE CONCENTRATION IS NOT KNOWN, THE HELICOPTER MUST BE DEFUELED AND REFUELED WITH FUEL CONTAINING THE PROPER ADDITIVE.

The required additive concentrations are in Section 2, Fuel limitations.

Blending the additive with the fuel is to be carried out using equipment normally supplied by the additive manufacturer. Follow manufacturer's recommended procedure when operating the blending equipment.

The flow of the additive must commence simultaneously with the flow of fuel. Refueling flow rates are to be between 95 and 227 liters per minute (25 and 60 US gal/min).

A differential refractometer may be used to accurately measure the concentration of additive in the fuel.

8.4.3 Mixing Alternate Fuel

CAUTION JP-4 OR COMMERCIAL JET B FUELS SHALL NOT BE MIXED WITH AVGAS.

As an alternate cold weather fuel when JP-4 or Commercial Jet B fuels cannot be obtained, use the following mixture:

- One part by volume AVGAS grade 80/87 or AVGAS 100L and,
- Two parts by volume Aviation Kerosene; JP-5; Commercial Jet A; or Commercial Jet A-1.

Refer to Fuel limitations in Section 2.

NOTE There is no engine operational time limit when using the AVGAS/Jet fuel mixture as long as 80/87 grade AVGAS is used and the 1:2 ratio is followed.

When operating on AVGAS/Jet fuel mixture, the anti-icing additive is required only for the jet fuel portion of the mixture.

CAUTION WHEN MIXING FUELS AVOID HIGH FLOW RATES. DO NOT EXCEED A FLOW RATE OF 189 LITERS PER MINUTE (50 US GAL/MIN). ALSO MAKE SURE THE FUEL NOZZLE AND FUEL TANK ARE ELECTRICALLY GROUNDED TO THE HELICOPTER.

Observe the following guidelines when making the alternate fuel mixture:

- Make the fuel mix in the helicopter tank. The AVGAS can be added before or after the jet fuel, the only restriction is that the final mix (including fuel in the tank before starting the mix) be two parts by volume jet fuel to one part by volume AVGAS.
- When the AVGAS/Jet fuel mixture is used in the engine the lead from the gasoline accumulates on the turbine and exhaust collector outlet ducts. The normal appearance after operation on lead fuel is a pale yellow powder deposit on the exhaust surfaces.
- If low temperature starting problems are encountered and the AVGAS/Jet fuel mixture is made in the fuel tank it may be necessary to bleed the unmixed fuel from the fuel lines in order to obtain a light-off. Refer to Allison Operation and Maintenance Manual

8.4.4 Oils

CAUTION DO NOT MIX MIL-L-23699 AND MIL-L-7808 OILS.

When temperatures are below -40°C (-40°F) OAT or MIL-L-23699 oils are not available, use MIL-L-7808G and subsequent oils.

8.4.5 Preheating

WARNING DO NOT USE AN OPEN FLAME HEATER TO PREHEAT THE ENGINES OR BATTERY.

At temperatures below -18°C OAT, preheat the engines concentrating heat application on the fuel controls.

CAUTION WHEN ENGINE AND MAIN TRANSMISSION OIL TEMPERATURES ARE BELOW -25°C PREHEAT THE OILS TO A TEMPERATURE OF -25°C OR HIGHER.

At temperatures below -30°C OAT, preheat the cockpit to a temperature of -10°C or higher for a duration of at least 15 minutes.

8.5 MAINTENANCE

The component retirement life limitations in the Airworthiness Limitations Section of the BO 105 Maintenance Manual shall be complied with.

Refer to the BO 105 Maintenance Manual for information on the helicopter inspection periods and special inspections for helicopter systems.

